



جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of the effect of Sleeping on Academic Performance

Submitted by: Rahaf Osama/ 24-101220

05/04 /2026

Introduction:

Many students struggle with their academic performance without knowing the reason, but the truth is that sleeping has a powerful impact on their academic performance. Sleeping plays an important role of focusing, memorising and understanding. This report explains how sleeping is important for students.

Research Question:

Does sleeping really affect students' academic performance?

Hypothesis

Population of Interest:

Our target population is University students.

Sampling Method:

This study used a **convenience sampling method**. This approach was suitable because it allowed quick data collection and helped identify general trends about sleep and academic performance. Efforts were made to include different students to improve the variety of responses

Bias Identification:

Bias type is **self-report bias**. the survey used clear and neutral questions and was anonymous to encourage honesty. Another bias is **sampling bias**, which was minimized by sharing the survey with different students.

Survey Questions:

- **How many hours do you usually sleep per night?**
- **Do you believe that your sleep affects your academic performance?**
- **How does your sleep schedule affect your academic performance?**
- **If you have not finished studying before an exam, what would you most likely do?**
- **What is the main reason for your lack of sleep?**
- **How would you describe your overall academic performance?**

Online survey link: https://docs.google.com/forms/d/1_x74Mhx5sbat6-DD6NnA96wUro0ALq2X_TZOoPx7fjI/edit

Number of samples collected: **75**

Analysis:

This study is made to show how the sleeping hours affects the academic performance of students, As There is a strong influence of sleeping on academic performance. In the previous survey, we asked students some questions about how many hours do they sleep daily and what are the main issues that affects their sleeping schedule and if this affects their academic performance or not. Th calculations are made as follows:

Mean of Sleep Hours = 7.05555555555555

Median of Sleep Hours = 7.5

Mode of Sleep Hours = 9.0

Mean of Academic Effect = 2.76388888888889

Median of Academic Effect = 3.0

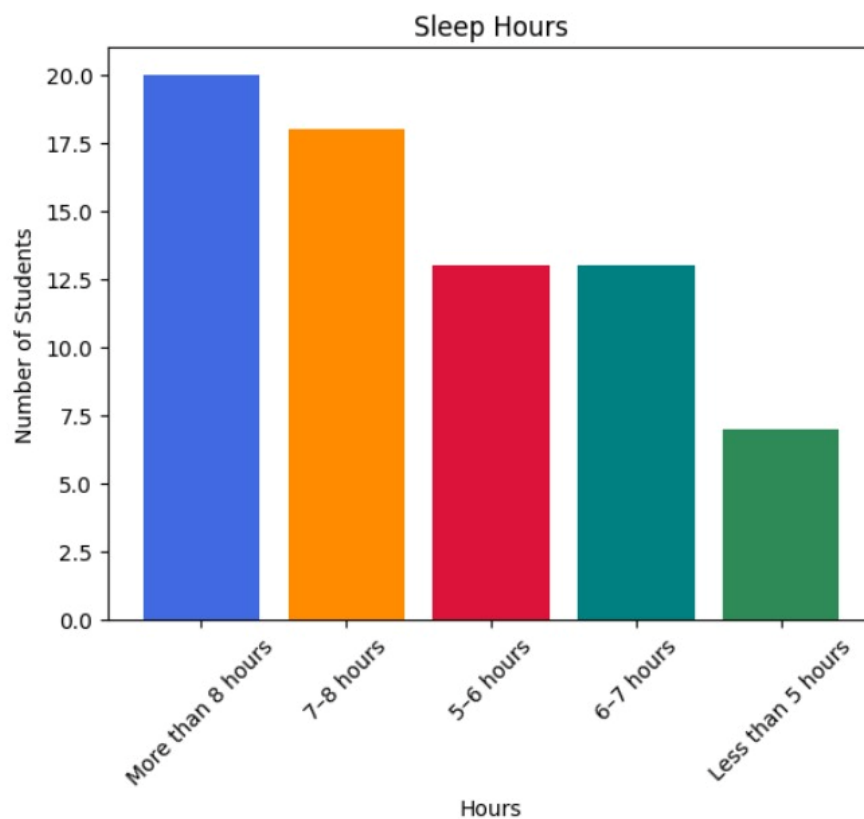
Mode of Academic Effect = 3. We expressed this analysis using statistical methods to show how students are affected by this relationship, as we represented it **graphically** and **numerically** as follows:

First question:

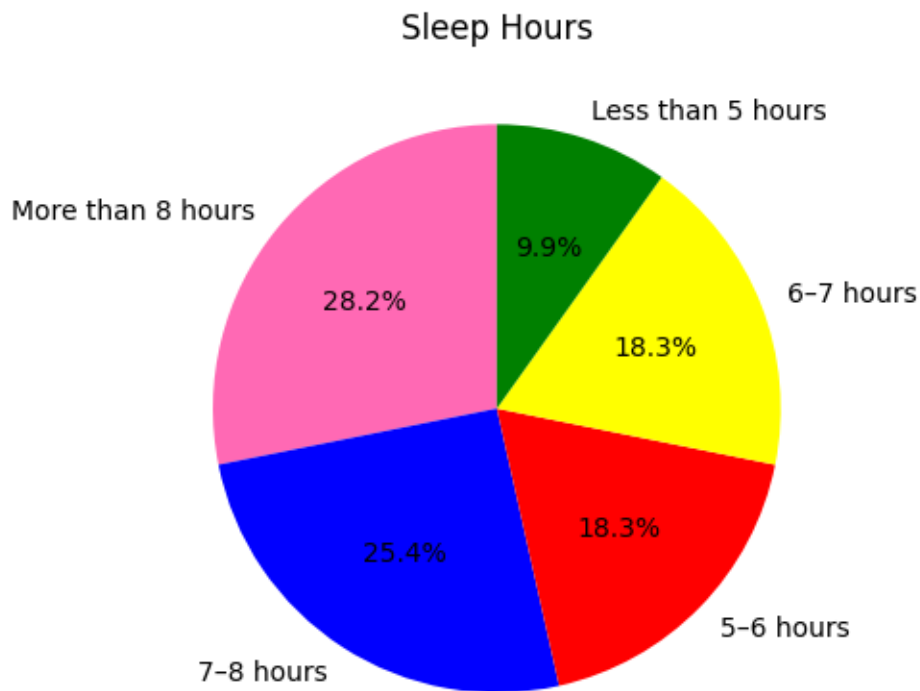
Numerically:

...	count
How many hours do you usually sleep per night?	
More than 8 hours	20
7–8 hours	18
5–6 hours	13
6–7 hours	13
Less than 5 hours	7
dtype: int64	

Graphically (by bar chart):



Graphically (by Pie chart):



According to the previous data representation, the total number of students responds to this question is **71 students**. The highest number of students usually **sleep more than 8 hours**. The number of students of the first category reaches **20 students (represents 28.2%)**, The number of students who usually sleep **7-8 hours** reaches **18 students (represents 25.4%)**. The number of students who usually **sleep 6-7 hours and 5-6 hours are equal** and each of them reaches **13 students (each represents 18.3%)**. The least number of students who usually **sleep less than 5 hours**, as it reaches **7 students (represents 9.9%)**. This shows that **most students prefer sleeping more hours to concentrate better**.

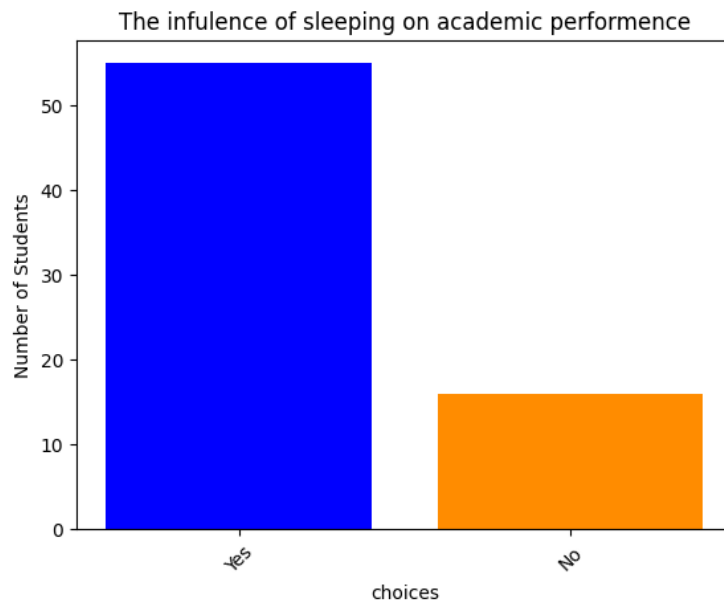
Second question:

Numerically:

		count
Do you believe that your sleep affects your academic performance?		
Yes		55
No		16

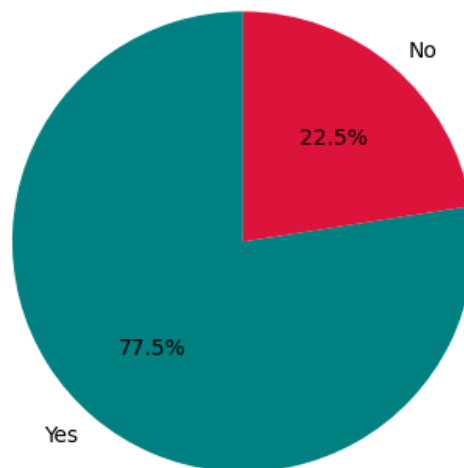
dtype: int64

Graphically (by bar chart):



Graphically (by Pie charts):

Sleep Effect on Academic Performance



Upon the previous data representation, the total number of responds on the second question is also **71 students**. The largest number of students believes that sleeping has influence on their academic performance as their number reaches **55 students (represents 77.5%)**, on the other hand, the other students believe that sleep doesn't have influence on their academic performance as their number reaches **16 students (represents 22.5%)**. **This shows that there is a relationship between sleeping and academic performance and sleeping has an influence on academic performance.**

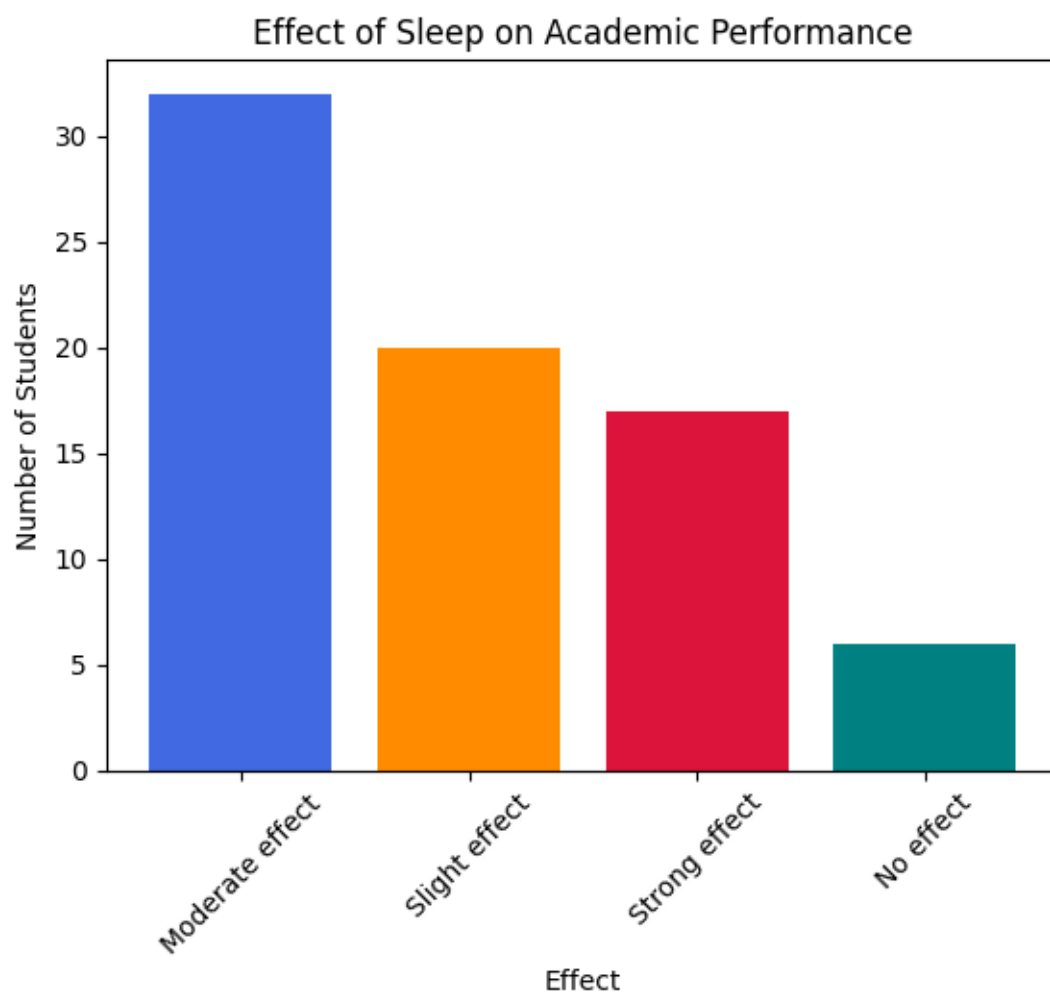
Third question:

Numerically:

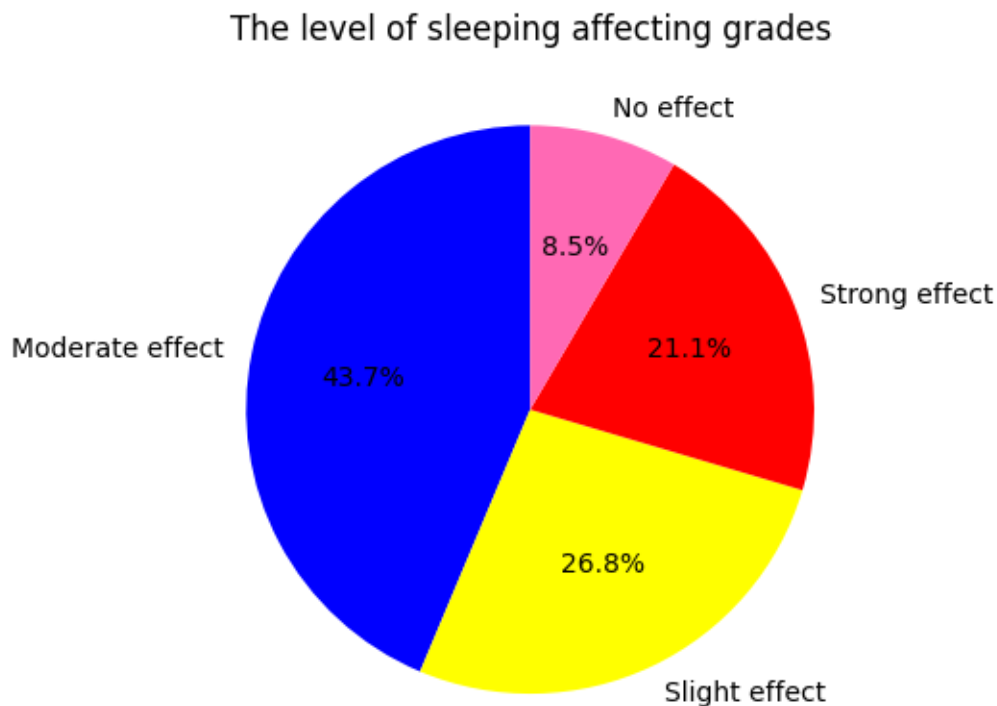
	count
How does your sleep schedule affect your academic performance (grades)?	
Moderate effect	31
Slight effect	19
Strong effect	15
No effect	6

dtype: int64

Graphically (by bar chart):



Graphically (by Pie charts):



Upon the previous data representation, the total number of students responds this question is **71 students**. The largest number of students believes that sleeping affects their academic performance **moderately** as their number reaches **31 students (represents 43.7%)**. The number of students who believe that sleeping affects them **slightly** reaches **19 students (represents 26.8%)**. The number of students who believe that sleeping affects their academic performance **strongly** reaches **15 students (represents 21.1%)**. The least number of students who believe that sleeping **doesn't affect** their academic performance reaches **6 students (represents 8.5%)**. This shows that most students believe that sleep has a slight to strong effect on their academic performance.

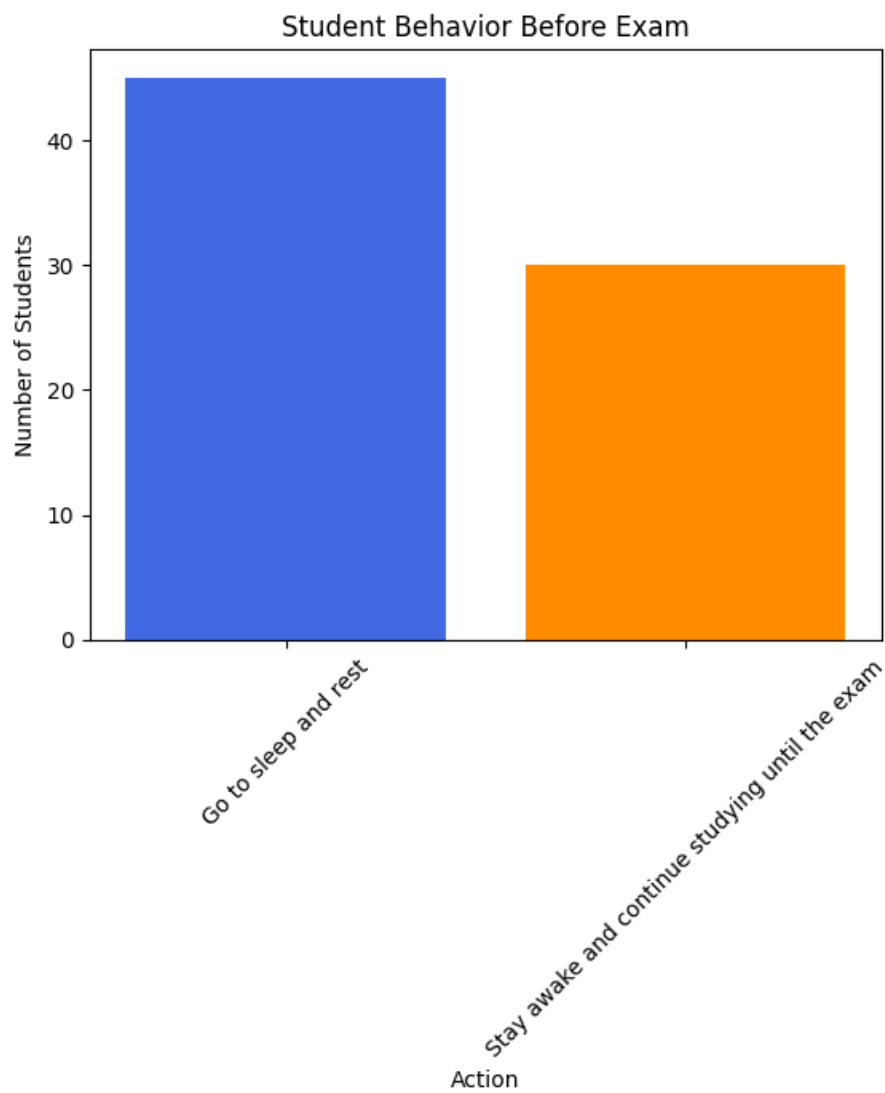
Fourth question:

Numerically:

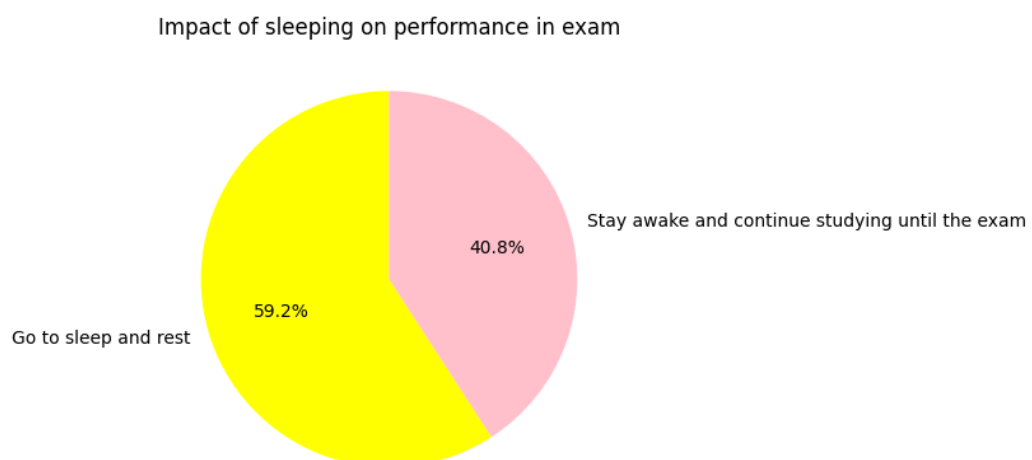
	count
If you have not finished studying before an exam, what would you most likely do?	
Go to sleep and rest	45
Stay awake and continue studying until the exam	30

dtype: int64

Graphically (by bar charts):



Graphically (by Pie charts):



Upon the previous data representation, the total number of students who responded to this question is **75 students**. The largest number of students believes that going to sleep and have a rest before exam as their number reaches **45 students (represents 59.2%)**, while the number of students who believes that staying awake and continue studying is the least as their number reaches **30 students (represents 40.8%)**. This shows that students prefer sleeping enough hours on important days more than staying up awake till the exam to keep themselves more focused and higher their concentration.

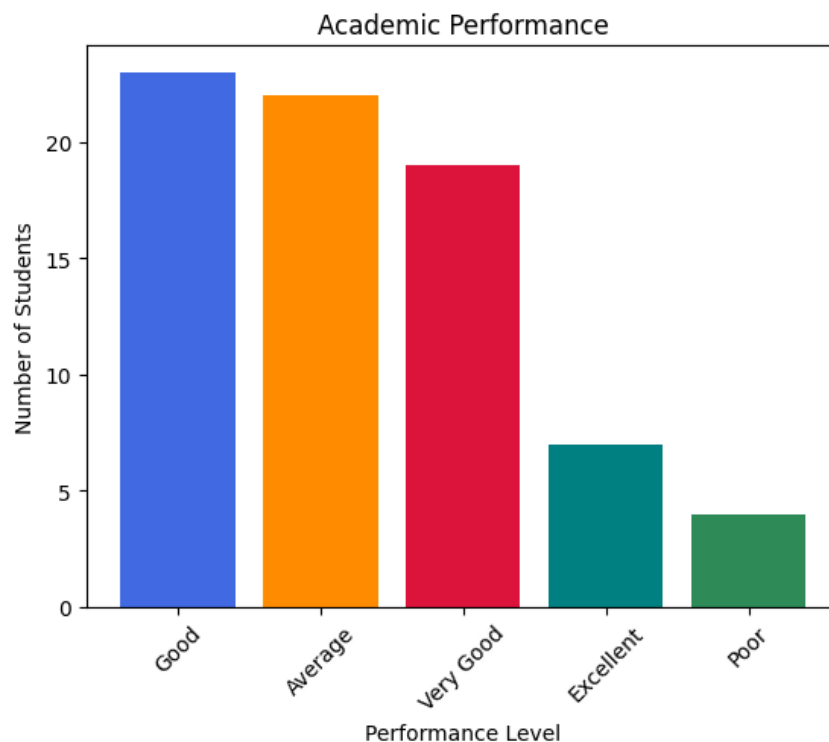
Fifth question:

Numerically:

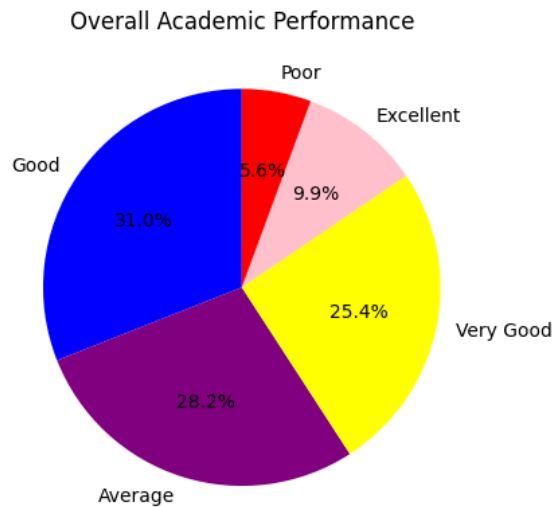
How would you describe your overall academic performance?		count
Good		23
Average		22
Very Good		19
Excellent		7
Poor		4

dtype: int64

Graphically (by bar chart):



Graphically (by Pie chart):

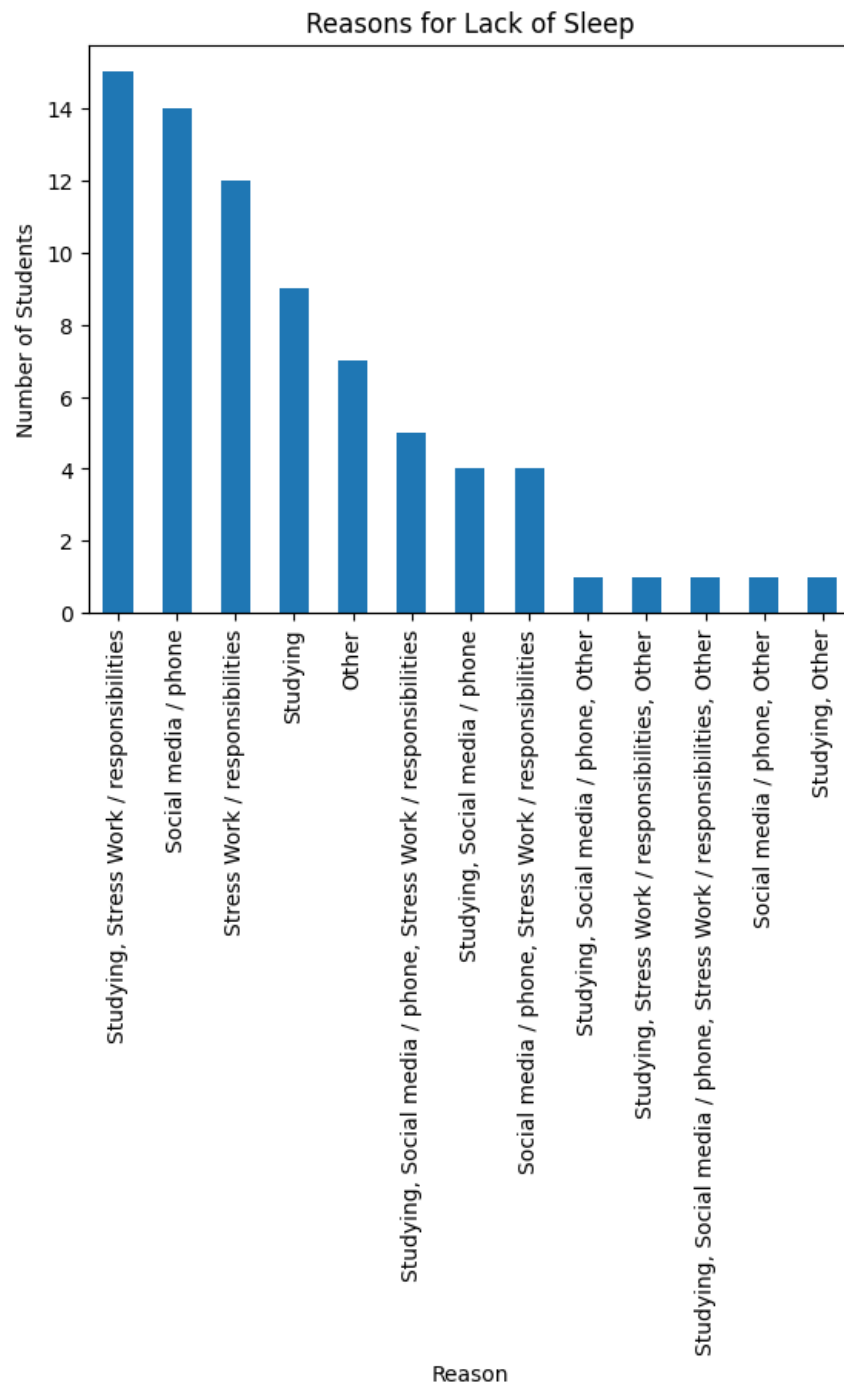


Upon the previous data representation, the total number of students who responded to this question is **75 students**. The largest number of students their academic performance is **Good** as their number reaches **23 students (represents 31.0%)**. The number of students whose academic performance is **Average** reaches **22 students (represents 28.2%)**. The number of students whose academic performance is **Very Good** reaches **19 students (represents 25.4%)**. The number of students whose academic performance is **Excellent** reaches **7 students (represents 9.9%)**. The least number of students whose academic performance is **Poor** reaches **4 students (represents 5.6%)**. This shows that how sleep patterns influence students' academic performances.

Sixth question: Numerically:

Studying, Stress Work / responsibilities	15
Social media / phone	14
Stress Work / responsibilities	12
Studying	9
Other	7
Studying, Social media / phone, Stress Work / responsibilities	5
Studying, Social media / phone	4
Social media / phone, Stress Work / responsibilities	4
Studying, Social media / phone, Other	1
Studying, Stress Work / responsibilities, Other	1
Studying, Social media / phone, Stress Work / responsibilities, Other	1
Social media / phone, Other	1
Studying, Other	1

Graphically (by bar chart):



Upon the previous data representation, the total number of students who responded to this question is **72 students**. The largest number of students believe that the studying, stress work and responsibilities are main reason for lack of sleep is **15 students**. The number of students believe that **social media and phone** are main reason for lack of sleep is **14 students**. The number of students believe that **studying only** is main reason for lack of sleep is 9 students. believe that **other** are main reason for lack of sleep is **7 students**. The number of students believe that **Studying, social media / phone, Stress Work / responsibilities** are main reason for lack of sleep is **5 students**. The other responses are shown in the previous data representation. **This shows that the main reason for students for getting less sleep is stress work , studying and responsibilities**

Conclusion

Eventually, the results of this study suggest a relationship between sleep and academic performance. The relationship between them appears to be directly proportional, as students who get sufficient hours of sleep tend to achieve better grades and higher academic performance. The findings indicate that adequate sleep may contribute positively to students' concentration, memory, and overall learning ability, which can lead to improved academic outcomes. However, while sleep plays an important role, it is not the only factor affecting performance, as other factors such as study habits, stress, and time management may also influence students' results